

Optimality Conditions

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When studying optimality conditions, we often discuss well-known conditions such as the Karush-Kuhn-Tucker (KKT) conditions and Fritz John (FJ) conditions. However, to build intuition, it is helpful to start with a simpler case: unconstrained optimization problems, and derive its necessary and sufficient conditions for optimality.

1 Optimality Conditions for Unconstrained Optimization

In this section, we consider the optimization problem where $X \subseteq \mathbb{R}^n$:

$$\min_x f(x) ; \quad x \in X. \quad (1.1)$$

In addition, we often make some basic assumptions:

- f is continuously differentiable (for first-order condition).
- f is twice continuously differentiable (for second-order condition).

The definition of local and global minimum:

- x^* is a *local minimum* if there exists an $\varepsilon > 0$ such that $f(x^*) \leq f(x)$, for all x with $\|x - x^*\| < \varepsilon$.
- x^* is a *global minimum* if $f(x^*) \leq f(x)$, for all $x \in X$.

We say a local or global minimum is *strict* if the corresponding inequality is strict for $x \neq x^*$.

1.1 Necessary Conditions for Optimality

If the objective function is differentiable, we can use gradients and Taylor approximations to compare the function value x^* with the cost of its close neighbors.

Theorem 1.1: First-order Necessary Condition

Let x^* be a local minimum of Problem 1.1, and suppose that f is continuously differentiable at x^* . Then

$$\nabla f(x^*) = 0.$$

Proof. Given some direction $d \in \mathbb{R}^n$ and scalar α , we have the first-order approximation

$$f(x^* + \alpha d) \approx f(x^*) + \nabla f(x^*)^\top ((x^* + \alpha d) - x^*).$$

Since x^* is a local minimum, we have the inequality

$$f(x^*) \leq f(x^*) + \nabla f(x^*)^\top (\alpha d).$$

For $\alpha > 0$ and $\alpha < 0$, we have $\nabla f(x^*)^\top d \geq 0$ for all d and $\nabla f(x^*)^\top d \leq 0$ for all d , respectively. Altogether, $\nabla f(x^*)^\top d = 0$ for all d , thus $\nabla f(x^*) = 0$. $\square$

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Similar to the first-order condition, we could also derive a second-order condition using second-order approximation.

Theorem 1.2: Second-order Necessary Condition

Let x^* be a local minimum of Problem 1.1, and suppose that f is twice continuously differentiable at x^* . Then

$$\nabla^2 f(x^*) \succeq 0.$$

Proof. Given some direction $d \in \mathbb{R}^n$ and scalar α , we have the second-order approximation

$$f(x^* + \alpha d) \approx f(x^*) + \alpha \nabla f(x^*)^\top d + \frac{\alpha^2}{2} d^\top \nabla^2 f(x^*) d.$$

Since x^* is a local minimum, we have the inequality

$$f(x^*) \leq f(x^*) + \alpha \nabla f(x^*)^\top d + \frac{\alpha^2}{2} d^\top \nabla^2 f(x^*) d.$$

By Theorem 1.1 and $\alpha^2 \geq 0$, we have $d^\top \nabla^2 f(x^*) d \geq 0$ for all d , i.e., $\nabla^2 f(x^*) \succeq 0$. $\square$

1.2 Sufficient Conditions for Optimality

Theorem 1.3: Second-order Sufficient Condition

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be twice continuously differentiable. Suppose that a vector x^* satisfies the conditions

$$\nabla f(x^*) = 0 \quad \text{and} \quad \nabla^2 f(x^*) \succ 0.$$

Then x^* is a strict local minimum of f . In particular, there exists scalars $\gamma > 0$ and $\varepsilon > 0$ such that

$$f(x) \geq f(x^*) + \frac{\gamma}{2} \|x - x^*\|^2,$$

for all x with $\|x - x^*\| \leq \varepsilon$.

Proof. Given some direction $d \in \mathbb{R}^n$, by the first condition and second-order approximation,

$$\begin{aligned} f(x^* + d) &= f(x^*) + \nabla f(x^*)^\top d + \frac{1}{2} d^\top \nabla^2 f(x^*) d + \mathcal{O}(\|d\|^2) \\ &= f(x^*) + \frac{1}{2} d^\top \nabla^2 f(x^*) d + \mathcal{O}(\|d\|^2). \end{aligned}$$

Since $\nabla^2 f(x^*)$ is real symmetric, we have the spectral decomposition $\nabla^2 f(x^*) = Q\Lambda Q^\top$, where Λ is a diagonal matrix of eigenvalues and Q is orthogonal whose columns are the corresponding eigenvectors. Let $z = Q^\top d$, we have $\|z\| = \|Q^\top d\| = \|d\|$ since Q is orthogonal and

$$d^\top \nabla^2 f(x^*) d = d^\top Q\Lambda Q^\top d = z^\top \Lambda z = \sum_{i=1}^n \lambda_i z_i^2 \geq \lambda_{\min} \sum_{i=1}^n z_i^2 = \lambda_{\min} \|z\|^2 = \lambda_{\min} \|d\|^2.$$

Substituting this inequality back to the second-order approximation, we have

$$\begin{aligned} f(x^* + d) &= f(x^*) + \frac{1}{2} d^\top \nabla^2 f(x^*) d + \mathcal{O}(\|d\|^2) \\ &\geq f(x^*) + \frac{\lambda_{\min}}{2} \|d\|^2 + \mathcal{O}(\|d\|^2) \\ &= f(x^*) + \left(\frac{\lambda_{\min}}{2} + \frac{\mathcal{O}(\|d\|^2)}{\|d\|^2} \right) \|d\|^2 \end{aligned}$$

Therefore, take $\gamma \leq \lambda_{\min} + 2\mathcal{O}(\|d\|^2)/\|d\|^2$ and $\|d\| = \|x - x^*\| \leq \varepsilon$, the proof is done. $\square$

1.3 Existence of Optimal Solutions

In general, a minimum need not exist. For example, $f(x) = x$ and $f(x) = e^x$ have no global minimum over $\mathbb{R}$. To show existence, we first review some definitions. Let $X \subseteq \mathbb{R}^n$,

- f is continuous at x if $\lim_{y \rightarrow x} f(y) = f(x)$.
- f is continuous on X if f is continuous at every $x \in X$.
- $f : X \rightarrow \mathbb{R}$ is lower semi-continuous at $x \in X$ if $f(x) \leq \liminf_{k \rightarrow \infty} f(x_k)$ for every sequence $\{x_k\}$ of X converging to x .
- $f : X \rightarrow \mathbb{R} \cup \{\pm\infty\}$ (extended valued function) is coercive if $\lim_{x \rightarrow \infty} f(x) = \infty$.

Proposition 1.4: Weierstrass' Theorem

Let $X \subseteq \mathbb{R}^n$ be nonempty and let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be lower semi-continuous at all points of X . Suppose that one of the following conditions holds:

1. X is compact (close and bounded).
2. X is closed and f is coercive.
3. There exists a scalar γ such that the level set $\{x \in X \mid f(x) \leq \gamma\}$ is nonempty and compact.

Then there exists a vector $x^* \in X$ such that $f(x^*) = \inf_{x \in X} f(x)$, i.e., x^* is a global minimum.

This can be reduced to the well-known *Weierstrass Extreme Value Theorem*, which states that every continuous function on a nonempty compact set attains its extreme values on that set, including a global minimum. We next give an example of using optimality conditions to prove a well-known inequality.

Example 1.5: Arithmetic-Geometric Mean Inequality

We show the arithmetic-geometric mean inequality:

$$\sqrt[n]{x_1 x_2 \cdots x_n} \leq \frac{x_1 + x_2 + \cdots + x_n}{n}.$$

Let $y_i = \ln(x_i)$ for all i , we can rewrite the inequality as

$$e^{y_1} + e^{y_2} + \cdots + e^{y_n} \geq n e^{\frac{y_1 + y_2 + \cdots + y_n}{n}}.$$

Let $y_1 + y_2 + \cdots + y_n = s$ and consider the optimization problem

$$\begin{aligned} \min_{y_i} \quad & e^{y_1} + e^{y_2} + \cdots + e^{y_n} \\ \text{s.t.} \quad & y_1 + y_2 + \cdots + y_n = s, \end{aligned}$$

we aim to show the optimal value is $ne^{s/n}$. We rewrite an equivalent unconstrained problem

$$\min_{y_i} f(y_1, y_2, \dots, y_{n-1}) = e^{y_1} + \cdots + e^{y_{n-1}} + e^{s - y_1 - \cdots - y_{n-1}}.$$

Note that since f is coercive, by Proposition 1.4, there exists a global minimum. Let $(y_1^*, y_2^*, \dots, y_{n-1}^*)$ be the global minimum, by Theorem 1.1, we have

$$\frac{\partial f}{\partial y_i} = e^{y_i} + e^{s-y_1-\dots-y_{n-1}}(-1) = 0 \text{ for } i = 1, \dots, n-1,$$

which implies $y_i^* = s - y_1 - \dots - y_{n-1}$ for $i = 1, \dots, n-1$. The system has only one solution: $y_i = s/n$ for all i , which is also the unique global minimum. Also, $e^{y_1^*} + \dots + e^{y_n^*} = ne^{s/n}$.

2 Lagrange Duality

In the previous section, we derived optimality conditions for unconstrained problems. Now we turn to constrained problems and introduce optimality conditions such as the Fritz John (FJ) and Karush-Kuhn-Tucker (KKT) conditions. However, since it is easier to understand these conditions with a basic knowledge of Lagrange duality, we first give a brief introduction to duality theory.

2.1 Lagrange Dual Problem

Consider the optimization problem

$$\begin{aligned} \min_x \quad & f_0(x) \\ \text{s.t.} \quad & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & h_i(x) = 0, \quad i = 1, \dots, l. \end{aligned} \tag{2.1}$$

with domain $\mathcal{D} = \bigcap_{i=1}^m \text{dom } f_i \cap \bigcap_{i=1}^l \text{dom } h_i$, and denote the optimal value by p^* . The idea of Lagrangian duality is to leverage the constraints in Problem (2.1) into the objective function, and thereby transforms a constrained problem to an unconstrained problem. We define the *Lagrangian* $\mathcal{L} : \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^l \rightarrow \mathbb{R}$ by

$$\mathcal{L}(x, \lambda, \nu) = f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \sum_{i=1}^l \lambda_i h_i(x),$$

where λ and ν are called the *Lagrange multipliers* or dual variables. We then define the (Lagrange) *dual function* $g : \mathbb{R}^m \times \mathbb{R}^l \rightarrow \mathbb{R}$ by

$$g(\lambda, \nu) = \inf_{x \in \mathcal{D}} \mathcal{L}(x, \lambda, \nu) = \inf_{x \in \mathcal{D}} \left(f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \sum_{i=1}^l \mu_i h_i(x) \right).$$

Note that the dual function is the point-wise infimum of a family of affine functions of (λ, ν) , which is always concave even if Problem (2.1) is not convex. Now suppose that x^* is an optimal solution, for any $\lambda \geq 0$ and any ν , we have

$$\mathcal{L}(x^*, \lambda, \nu) = f_0(x^*) + \underbrace{\sum_{i=1}^m \lambda_i f_i(x^*)}_{\leq 0} + \underbrace{\sum_{i=1}^l \lambda_i h_i(x^*)}_{=0} \leq f_0(x^*),$$

which implies

$$g(\lambda, \nu) = \inf_{x \in \mathcal{D}} \mathcal{L}(x, \lambda, \nu) \leq \inf_{x \in \mathcal{D}} \mathcal{L}(x^*, \lambda, \nu) \leq f_0(x^*) = p^*. \tag{2.2}$$

That is, the optimal value p^* is an upper bound of the dual function g , which is also the main idea of *weak duality*. Also, the optimal value p^* is a lower bound of the objective function f_0 . Figure 2.1 illustrates this property.

Example 2.1

Consider the optimization problem

$$\begin{aligned} \min_x \quad & x^3 + 2x^2 - x + 1 \\ \text{s.t.} \quad & x^2 \leq 1 \iff 0 \leq x \leq 1. \end{aligned}$$

The dual function is

$$g(\lambda) = \inf_x (x^3 + 2x^2 - x + 1 + \lambda(x^2 - 1)).$$

We can observe from Figure 2.1 that the optimal value is the lower bound of the objective function, and also is the upper bound of the dual function.

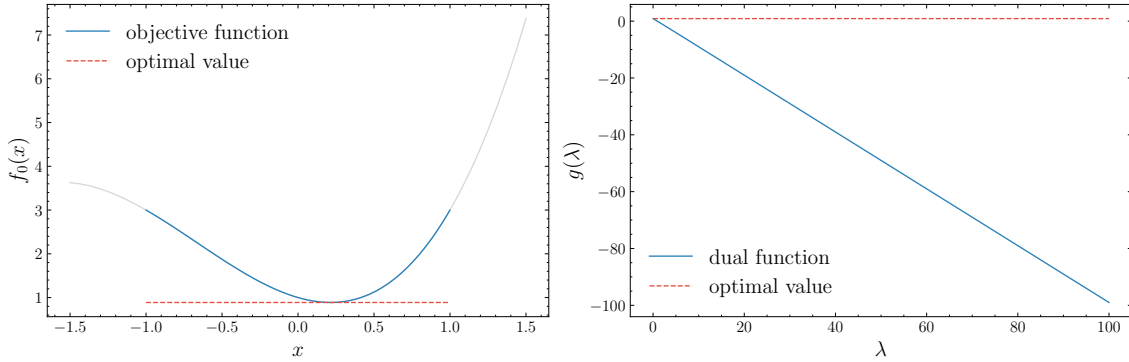


Figure 2.1: Lower and upper bound in Example 2.1.

Since the optimal value is an upper bound of the dual function, the (Lagrange) *dual problem* is formulated by maximizing the dual function. Formally, we define the dual problem as

$$\begin{aligned} \max_{\lambda, \nu} \quad & g(\lambda, \nu) \\ \text{s.t.} \quad & \lambda \succeq 0. \end{aligned} \tag{2.3}$$

The dual problem is always convex whether the primal problem is convex or not.

2.2 Weak Duality and Strong Duality

Denote d^* as the optimal value of the dual problem. Since the optimal value of the primal problem p^* is an upper bound of the dual function, we have the *weak duality*:

$$d^* \leq p^*.$$

We define $p^* - d^*$ as the duality gap. We say that *strong duality* holds if

$$d^* = p^*.$$

While strong duality does not hold in general, there are some results that establish conditions on the problem under which strong duality holds for *convex optimization*. These conditions are called *constraint qualifications*. Here we give a simple and widely used constraint qualification in the context of convex optimization, the *Slater's condition*: There exists an $x \in \text{relint}\mathcal{D}$ such that $f_i(x) < 0$ for all $i = 1, \dots, m$ and $h_i(x) = 0$ for all $i = 1, \dots, l$, i.e., there exists a *strictly feasible* point in the relative interior of the feasible set.

2.3 Complementary Slackness

Suppose that the primal optimal value p^* and dual optimal value d^* are attained where x^* is the primal optimal solution and (λ^*, ν^*) is the dual optimal solution. This means that

$$g(\lambda^*, \nu^*) \leq f_0(x^*) + \underbrace{\sum_{i=1}^m \lambda_i^* f_i(x^*)}_{\leq 0} + \underbrace{\sum_{i=1}^l \nu_i^* h_i(x^*)}_{=0} \leq f_0(x^*).$$

The first inequality is because infimum is a lower bound, and the second inequality is because x^* is feasible such that $f_i(x^*) \leq 0$ for $i = 1, \dots, m$ and $h_i(x^*) = 0$ for $i = 1, \dots, l$. Now suppose that strong duality holds, i.e., $g(\lambda^*, \nu^*) = f_0(x^*)$, we have that the two inequalities in the chain hold equality. This implies the *complementary slackness*:

$$\lambda_i^* f_i(x^*) = 0, \quad i = 1, \dots, m. \quad (2.4)$$

We say an inequality constraint is *binding* or *active* if $f_i(x^*) = 0$. An important result of complementary slackness is that the i -th inequality constraint is binding if λ_i^* is not zero, or λ_i^* is zero if the i -th inequality constraint is non-binding.

- $\lambda_i^* > 0 \implies f_i(x^*) = 0$.
- $f_i(x^*) < 0 \implies \lambda_i^* = 0$.

Roughly speaking, only the constraints that are binding at the optimal point have a direct impact on the optimal solution through their Lagrange multipliers, while constraints that are non-binding do not affect the solution, as their Lagrange multipliers are zero. Complementary slackness allows us to identify which constraints are active and further simplify an optimization problem, or identify which constraints are critical and provide insights for decision-making.

Another result of complementary slackness is that since

$$f_0(x^*) = g(\lambda^*, \nu^*) \leq f_0(x^*) + \sum_{i=1}^m \lambda_i^* f_i(x^*) + \sum_{i=1}^l \nu_i^* h_i(x^*) = \mathcal{L}(x^*, \lambda^*, \nu^*),$$

we can conclude that x^* minimizes $\mathcal{L}(x, \lambda^*, \nu^*)$. Therefore, by Theorem 1.1, it follows that

$$\nabla f_0(x^*) + \sum_{i=1}^m \lambda_i^* \nabla f_i(x^*) + \sum_{i=1}^l \nu_i^* \nabla h_i(x^*) = 0. \quad (2.5)$$

Equation (2.5) is also known as the *stationary condition*.

3 Fritz John Conditions

In this section, we state the Fritz John (FJ) necessary and sufficient conditions. These necessary conditions, including the FJ necessary conditions and KKT necessary conditions, basically involves a stationary condition, complementary slackness, primal feasibility, and dual feasibility. These conditions rely on different assumptions, which involves continuity, differentiability, and convexity on objective functions, binding/non-binding inequality constraints, and equality constraints. Thereby, our focus will be on clearly stating the assumptions behind each condition and providing intuitive explanations rather than full proofs.

We begin by introducing two cones at a given point x , the *cone of feasible directions* $F_X(x) = \{d \mid x + \alpha d \in X\}$ and the *cone of descent directions* $D_X(x) = \{d \mid f_0(x + \alpha d) < f_0(x)\} = \{d \mid \nabla f_0(x)^\top d < 0\}$, both for all $\alpha \in (0, \bar{\alpha})$ for some $\bar{\alpha}$. This leads to the most important intuition:

$$x \text{ is a local minimum} \implies D_X(x) \cap F_X(x) = \emptyset,$$

a local minimum is attained when any descent direction is not a feasible direction (or vice versa).

3.1 Fritz John Necessary Conditions

The FJ necessary conditions if for local minimums. We need some assumptions:

- Objective function and binding inequality constraints are differentiable.
- Non-binding inequality constraints are continuous.
- Equality constraints are continuously differentiable.

Theorem 3.1: Fritz John Necessary Conditions

Let x be a feasible solution and $I = \{i \mid f_i(x) = 0\}$ be the set of binding constraints. Suppose that f_0, f_i for $i \in I$ are differentiable at x , f_i for $i \notin I$ are continuous at x , and h_i for $i = 1, \dots, l$ are continuously differentiable at x . If x is a local minimum, then there exists λ_0, λ_i for $i \in I$, and ν_i for $i = 1, \dots, l$ such that

1. Stationary condition: $\lambda_0 \nabla f_0(x) + \sum_{i \in I} \lambda_i \nabla f_i(x) + \sum_{i=1}^l \nu_i \nabla h_i(x) = 0$.
2. Primal feasibility: $f_i(x) \leq 0$ for $i = 1, \dots, m$ and $h_i(x) = 0$ for $i = 1, \dots, l$.
3. Dual feasibility: $\lambda_0, \lambda_i \geq 0$ for $i \in I$.
4. Lagrange multipliers are not all zero: $(\lambda_0, \lambda, \nu) \neq 0$.

To show this, we define another set $G_X(x) = \{d \mid \nabla f_i(x)^\top d < 0, i \in I\}$, which means that if we move by direction d , every binding constraints will still strictly satisfy. This set is like a strictly feasible set that is a subset of the cone of feasible directions, i.e., $G_X(x) \subseteq F_X(x)$, which leads to the following result:

$$D_X(x) \cap F_X(x) = \emptyset \implies D_X(x) \cap G_X(x) = \emptyset.$$

Now if x is a local minimum, there exists no direction d that satisfies both $\nabla f_0(x)^\top d < 0$ and $\nabla f_i(x)^\top d < 0$ for all $i \in I$. Let $\nabla f_0(x)^\top, \nabla f_i(x)^\top, i \in I$ be the rows of a matrix A , we have $Ad < 0$ has no solution. Then by Gordan's Lemma, the system $A^\top \lambda = 0, \lambda \geq 0$ for some non-zero λ has a solution. Take $\lambda = (\lambda_0, \lambda_i, \dots)$ for $i \in I$ as the Lagrange multipliers, $A^\top \lambda = 0$ is the stationary condition, $\lambda \geq 0$ is the dual feasibility, and the Lagrange multipliers are not all zero. Seems like the proof is done, however, this is actually just for problems with inequality constraints only. We need an additional set for equality constraints and the full proof is more tedious, please refer to Bazaraa [1] for the full proof.

Theorem 3.1 can be reduced to a more used FJ necessary conditions [4] with complementary slackness by imposing stronger assumptions:

- Objective function and inequality constraints are differentiable.
- Equality constraints are continuously differentiable.

Corollary 3.2

Let x be a feasible solution. Suppose that f_0, f_i for $i = 1, \dots, m$ are differentiable at x , and h_i for $i = 1, \dots, l$ are continuously differentiable at x . If x is a local minimum, then there exists λ_0, λ_i for $i = 1, \dots, m$, and ν_i for $i = 1, \dots, l$ such that

1. Stationary condition: $\lambda_0 \nabla f_0(x) + \sum_{i=1}^m \lambda_i \nabla f_i(x) + \sum_{i=1}^l \nu_i \nabla h_i(x) = 0$.
2. Complementary slackness: $\lambda_i f_i(x) = 0$ for $i = 1, \dots, m$.
3. Primal feasibility: $f_i(x) \leq 0$ for $i = 1, \dots, m$ and $h_i(x) = 0$ for $i = 1, \dots, l$.

4. Dual feasibility: $\lambda_0, \lambda_i \geq 0$ for $i = 1, \dots, m$.
5. Lagrange multipliers are not all zero: $(\lambda_0, \lambda, \nu) \neq 0$.

Since all non-binding constraints are now differentiable, one can extend $G_X(x)$ to all inequality constraints instead of binding constraints only). Then by setting the Lagrange multipliers for non-binding constraints to 0, i.e., $\lambda_i = 0$ for $i \notin I$, we obtain the result with the same procedure above. In addition, we say (x, λ, ν) is a FJ point if it satisfies the FJ necessary conditions.

3.2 Fritz John Sufficient Conditions

The FJ sufficient conditions is also for local minimums. Again, we need some assumptions:

- Objective function is pseudoconvex.
- Binding inequality constraints are strictly pseudoconvex.
- Equality constraints are affine.
- Gradient of equality constraints are linear independent.

Theorem 3.3: Fritz John Sufficient Conditions

Let $I = \{i \mid f_i(x) = 0\}$ be the set of binding constraints and define $S = \{x \mid f_i(x) \leq 0 \text{ for } i \in I, h_i(x) = 0 \text{ for } i = 1, \dots, l\}$. Suppose that x is a FJ point, h_i for $i = 1, \dots, l$ are affine, ∇h_i for $i = 1, \dots, l$ are linear independent. If there exists some neighborhood $N_\varepsilon(x)$ such that f_0 is pseudoconvex on $S \cap N_\varepsilon(x)$ and f_i for $i \in I$ are strictly pseudoconvex on $S \cap N_\varepsilon(x)$, then x is a local minimum.

4 Karush-Kuhn-Tucker Conditions

The FJ conditions provide a general set of necessary conditions for optimality, which does not require the non-negativity of the Lagrange multiplier associated with the objective function. This makes FJ conditions less restrictive, for example, if $\lambda_0 = 0$, the stationary condition no longer reflects a balance between the objective and the constraints. The main difference between the FJ conditions and the KKT conditions is that the Lagrange multiplier λ_0 cannot be zero, i.e., $\lambda_0 > 0$. Thus, the KKT conditions can be seen as a special case of the FJ conditions, where the presence of regularity in the problem ensures meaningful Lagrange multipliers. We can derive the KKT conditions from the FJ conditions if some *constraint qualification* holds. Although constraint qualifications were briefly introduced in Section 2.2, they are more broadly used to derive the KKT conditions and will be discussed in more detail in Section 5.

4.1 Karush-Kuhn-Tucker Necessary Conditions

Under appropriate assumptions, including a valid constraint qualification, the KKT necessary conditions hold for a local minimum.

- Objective function and binding inequality constraints are differentiable.
- Non-binding constraints are continuous.
- Equality constraints are continuously differentiable.
- Constraint qualification.

Theorem 4.1: Karush-Kuhn-Tucker Necessary Conditions

Let x be a feasible solution and $I = \{i \mid f_i(x) = 0\}$ be the set of binding constraints. Suppose that f_0, f_i for $i \in I$ are differentiable at x , f_i for $i \notin I$ are continuous at x , and h_i for $i = 1, \dots, l$ are continuously differentiable at x . Suppose some constraint qualification holds (e.g., $\nabla f_i(x)$ for $i \in I$ and $\nabla h_i(x)$ for $i = 1, \dots, l$ are linear independent). If x is a local solution, then there exists *unique* λ_i for $i \in I$, ν_i for $i = 1, \dots, l$ such that

1. Stationary condition: $\nabla f_0(x) + \sum_{i \in I} \lambda_i \nabla f_i(x) + \sum_{i=1}^l \nu_i \nabla h_i(x) = 0$.
2. Primal feasibility: $f_i(x) \leq 0$ for $i = 1, \dots, m$ and $h_i(x) = 0$ for $i = 1, \dots, l$.
3. Dual feasibility: $\lambda_i \geq 0$ for $i \in I$.

Recall that we derived FJ necessary conditions by $D_X(x) \cap G_X(x) = \emptyset$. Now suppose that $G_X(x) \neq \emptyset$. In this case, let $\nabla f_i(x)^\top$ for $i \in I$ be the rows of a matrix A , the system $Ad < 0$ has some solution. By Gordan's Lemma, the system $A^\top \lambda = 0$, $\lambda \geq 0$ for some non-zero λ has no solution. Note that this always holds true if $\nabla f_i(x)$ for $i \in I$ are *linear independent*. In this case, the FJ conditions' dual feasibility for $i \in I$ are not satisfied, we must have $\lambda_0 > 0$ for the dual feasibility to hold.

Again, this is for problems with inequality constraints only. Similar to the FJ necessary conditions, Theorem 4.1 can be reduced to a more used KKT necessary conditions with complementary slackness by assuming the differentiability of non-binding constraints:

- Objective function and inequality constraints are differentiable.
- Equality constraints are continuously differentiable.
- Constraint qualification.

Corollary 4.2

Let x be a feasible solution. Suppose that f_0, f_i for $i = 1, \dots, m$ are differentiable at x , and h_i for $i = 1, \dots, l$ are continuously differentiable at x . Suppose some constraint qualification holds (e.g., Slater's condition). If x is a local minimum, then there exists λ_0, λ_i for $i = 1, \dots, m$, and ν_i for $i = 1, \dots, l$ such that

1. Stationary condition: $\nabla f_0(x) + \sum_{i \in I} \lambda_i \nabla f_i(x) + \sum_{i=1}^l \nu_i \nabla h_i(x) = 0$.
2. Complementary slackness: $\lambda_i f_i(x) = 0$ for $i = 1, \dots, m$.
3. Primal feasibility: $f_i(x) \leq 0$ for $i = 1, \dots, m$ and $h_i(x) = 0$ for $i = 1, \dots, l$.
4. Dual feasibility: $\lambda_i \geq 0$ for $i = 1, \dots, m$.

We say (x, λ, ν) is a KKT point if it satisfies the KKT necessary conditions.

4.2 Karush-Kuhn-Tucker Sufficient Conditions

In a similar manner to the FJ sufficient conditions, the KKT sufficient conditions ensure global optimality when the problem satisfies specific convexity properties.

- Objective is pseudoconvex.
- Binding inequality constraints are quasiconvex.
- Equality constraints are quasiconvex/quasiconcave.

Theorem 4.3: Karush-Kuhn-Tucker Sufficient Conditions

Suppose that x is a KKT point, and let $I = \{i \mid f_i(x) = 0\}$ be the set of binding constraints. If f_0 is pseudoconvex, f_i for $i \in I$ are quasiconvex, and h_i is quasiconvex if $\nu_i > 0$ and quasiconcave if $\nu_i < 0$, then x is a global minimum.

For convex optimization problems, the KKT conditions are sufficient. Here is an example of solving a convex program with the KKT conditions.

Example 4.4

Consider the convex program

$$\begin{aligned} \min_x \quad & f_0(x_1, x_2) = (x_1 - 1)^2 + (x_2 - 2)^2 \\ \text{s.t.} \quad & f_1(x_1, x_2) = x_1 + 3x_2 - 1 \leq 0 \end{aligned}$$

We first check the Slater's condition. We have a feasible point $(0, 0)$ where $f_1(0, 0) = -1$ is strictly feasible (and both the objective and constraint are differentiable). Then by the KKT conditions, there exists a unique λ such that stationary condition holds

$$\nabla f_0(x_1, x_2) + \lambda \nabla f_1(x_1, x_2) = 0 \implies 2x_1 - 2 + \lambda = 0, \quad 2x_2 - 4 + 3\lambda = 0,$$

complementary slackness holds

$$\lambda f_1(x_1, x_2) = 0 \implies \lambda(x_1 + 3x_2 - 1) = 0,$$

and primal/dual feasibility holds. We first solve the system

$$\begin{cases} 2x_1 - 2 + \lambda = 0 \\ 2x_2 - 4 + 3\lambda = 0 \\ \lambda(x_1 + 3x_2 - 1) = 0, \end{cases}$$

then check the primal/dual feasibility. Suppose that the constraint is binding, the system becomes

$$\begin{cases} 2x_1 - 2 + \lambda = 0 \\ 2x_2 - 4 + 3\lambda = 0 \\ x_1 + 3x_2 - 1 = 0 \end{cases} \implies (x_1, x_2, \lambda) = \left(\frac{2}{5}, \frac{1}{5}, \frac{6}{5}\right),$$

where (x_1, x_2, λ) indeed satisfies primal and dual feasibility.

5 Constraint Qualifications

The FJ necessary conditions are a general set of conditions that hold for a local minimum. The KKT necessary conditions reduce the FJ conditions by requiring that the Lagrange multiplier associated to the objective function is positive, which requires a *constraint qualification* (regularity condition) to hold. In Section 4.1, we mentioned that the linear independence of gradients of binding constraints help guarantees the KKT necessary conditions from FJ necessary conditions. In this section, we introduce a few more constraint qualifications.

$$\text{local point (FJ point) + constraint qualification} \implies \text{KKT point.}$$

We first introduce another cone: the cone of tangents. A tangent cone $T_X(x)$ is the set of all tangents at point x , where a direction d is a tangent of x if there exists a sequence $\{x_k\} \subset X$

such that $d = \lim_{k \rightarrow \infty} \lambda_k(x_k - x)$ for all $\lambda_k > 0$ and $x_k \rightarrow x$. The tangent cone is generally larger than the cone of feasible directions, i.e., $F_X(x) \subseteq T_X(x)$. For example, consider $x_2 \geq x_1^2$ at point $(0,0)$, then the direction $d = (0,1)$ is a tangent but not a feasible direction. Instead of using the cone of feasible directions such that local minimum implies $D_X(x) \cap F_X(x) = \emptyset$, replacing $F_X(x)$ by the tangent cone achieves the same result [1, Theorem 5.1.2]:

$$x \text{ is a local minimum} \implies D_X(x) \cap T_X(x) = \emptyset.$$

In the following constraint qualifications, we only give the intuition of the proof for problems with inequality constraints only. Please refer to Bazaraa [1] for the full proof.

Theorem 5.1: Abadie Constraint Qualification

Suppose that x satisfies the FJ necessary conditions. Let $I = \{i \mid f_i(x) = 0\}$ be the set of binding constraints, if the constraint qualification:

$$\begin{aligned} T_X(x) &= G'_X(x) \cap H_X(x); \\ G'_X(x) &= \{d \mid \nabla g_i(x)^\top d \leq 0, \ i \in I\}, \\ H_X(x) &= \{d \mid \nabla h_i(x)^\top d = 0, \ i = 1, \dots, l\}, \end{aligned}$$

then x satisfies the KKT necessary conditions.

Since $T_X(x) = G'_X(x)$, we have $D_X(x) \cap G'_X(x) = \emptyset$. Let $-\nabla f_i(x)^\top$ for $i \in I$ be the rows of a matrix A , the system $Ad \geq 0$ and $\nabla f_0(x)^\top d < 0$ has no solution. By Farkas' Lemma, the system $A^\top \lambda = \nabla f_i(x)$ has a solution. Take $\lambda = (\lambda_i, \dots)$ for $i \in I$, we get the stationary condition with $\lambda_0 = 1$, therefore the KKT necessary conditions hold.

Theorem 5.2: Linearity Constraint Qualification (LCQ)

The statement of Theorem 5.1 holds true with the constraint qualification:

$$f_i, \ i = 1, \dots, m \text{ and } h_i, \ i = 1, \dots, l \text{ are all affine.}$$

Since f_i are linear, without loss of generality we write the inequality constraints as $Ax = b$. We next claim that *LCQ implies the Abadie constraint qualification*, i.e., $T_X(x) = G'_X(x)$. Let $d = \lim_{k \rightarrow \infty} \lambda_k(x_k - x) \in T_X(x)$, we have

$$A(x_k - x) = Ax_k - Ax \leq b - b = 0.$$

Multiplied by λ_k and take the limits, we have

$$A \lim_{k \rightarrow \infty} \lambda_k(x_k - x) = Ad \leq 0,$$

which implies $d \in G'_X(x)$ and thereby $T_X(x) \subseteq G'_X(x)$. On the other hand, let $d \in G'_X(x)$, i.e., $Ad \leq 0$, then $A(x + \alpha d) \leq b$ for some sufficiently small positive α . Non-binding constraints will also always hold true for some sufficiently small positive α . Thus, d is a feasible direction, i.e., $d \in F_X(x) \subseteq T_X(x)$ and thereby $G'_X(x) \subseteq T_X(x)$. Together, we have $T_X(x) = G'_X(x)$.

Theorem 5.3: Linear Independence Constraint Qualification (LICQ)

The statement of Theorem 5.1 holds true with the constraint qualification:

$$\nabla f_i(x), \ i \in I \text{ and } \nabla h_i(x), \ i = 1, \dots, l \text{ are linear independent.}$$

As shown in the proof of Theorem 4.1, if $\nabla f_i(x)$ for $i \in I$ are linear independent, let $\nabla f_i(x)^\top$ for $i \in I$ be the rows of a matrix A , then $A^\top \lambda = 0$ has only the trivial solution $\lambda = 0$. Thus the system $A^\top \lambda = 0$, $\lambda \geq 0$ for some non-zero λ has no solution. Since x satisfies the FJ necessary conditions, λ_0 cannot be zero (otherwise the dual feasibility would not hold), therefore $\lambda_0 > 0$ and the KKT necessary conditions hold.

Theorem 5.4: Mangasarian-Fromovitz Constraint Qualification (MFCQ)

The statement of Theorem 5.1 holds true with the constraint qualification:

there exists a vector d such that $\nabla f_i(x)^\top d < 0$, $i \in I$ and $\nabla h_i(x)^\top d = 0$,
and $\nabla h_i(x)$, $i = 1, \dots, l$ are linear independent.

Given the fact that there exists d such that $\nabla f_i(x)^\top d < 0$ for $i \in I$, the system $Ad < 0$ has a solution. Then by Gordan's Lemma, the system $A^\top \lambda = 0$, $\lambda \geq 0$ for some non-zero λ has no solution. It follows from the proof of Theorem 5.3 that the KKT necessary conditions hold.

In summary, constraint qualifications serve as the bridge between the generality of the FJ conditions and the stronger guarantees offered by the KKT conditions. For more information, please refer to Bazaraa [1].

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